

A scientist examining action potentials in mature muscle cells identifies certain muscle cells that, upon receiving an action potential, can directly transfer ions to adjacent cells. These cells would most likely include:

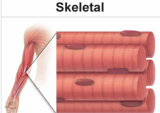
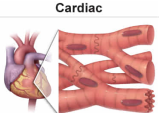
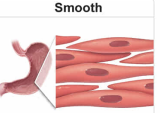
- I. smooth muscle cells.
- II. cardiac muscle cells.
- III. skeletal muscle cells.

- ☐ A. I only [4%]
- ☐ B. II only [32%]
- ✓ ☒ C. I and II only [34%]
- ☐ D. II and III only [28%]

Correct

34% answered correctly

Explanation:

Types of muscle cells (myocytes)			
	Skeletal	Cardiac	Smooth
			
Appearance	• Bunched fibers • Striated (has sarcomeres) • Multinucleated	• Branched fibers • Striated (has sarcomeres) • 1-2 nuclei	• Bunched cells • Not striated (no sarcomeres) • Uninucleated
Location & function	• Attached to bone via tendons & leverages bones to generate movement • Aids circulation by squeezing vessels • Heat production (shivering)	• Makes up heart walls & pumps blood through the circulatory system	• Lines hollow visceral organs (eg, stomach) & promotes substance (eg, food) movement • Lines blood vessels & affects blood pressure
Gap junctions	• Not present (cells are electrically isolated)	• Present (cells can transfer action potentials directly to adjacent cells)	
Control	• Voluntary • Somatic nervous system (ACh)	• Involuntary • Myogenic (particularly the heart) • Regulated by autonomic nervous system	

Myocytes (ie, muscle cells) generate forces that allow various functions to be performed in the body (eg, movement of limbs, pumping of blood, movement of food through the digestive tract). The three main types of myocytes are skeletal muscle cells, cardiac muscle cells, and smooth muscle cells, each of which exhibits specific characteristics:

- **Skeletal muscle cells** are long, **cylindrical fibers** that each possess **many nuclei**. Skeletal muscle cells contain **sarcomeres** and are therefore **striated** (ie, striped) in appearance. In addition, mature skeletal muscle cells *lack gap junctions* and are therefore **electrically isolated** from one another (ie, *cannot* directly pass ions/**action potentials** between cells).
- **Cardiac muscle cells** are short and **branched** cells with **one or two nuclei** each. These cells also possess **sarcomeres** that cause the cells to exhibit a

striated appearance. Cardiac muscle cells are connected to adjacent cells via **intercalated discs**, which contain desmosomes (providing strong adhesion between cells) and **gap junctions**, openings between adjacent cells that permit the passage of certain molecules (eg, ions) between cells.

- **Smooth muscle cells** are short cells that are wide in the middle and tapered (ie, narrower) at each end. Each smooth muscle cell possesses a **single nucleus**. These cells **lack sarcomeres** and are therefore **not striated** in appearance. In addition, many types of smooth muscle cells **possess gap junctions**.

In this question, a scientist identifies certain mature muscle cells that, upon receiving an action potential, can directly transfer ions to adjacent cells. Ions cannot **permeate** the lipid bilayer of **cell membranes** directly and therefore require openings (such as gap junctions) to pass between cells. Accordingly, these cells likely include both **smooth** and **cardiac muscle** cells, which are the *only* muscle cell types that possess gap junctions (**Numbers I and II**).

(Number III) Mature skeletal muscle cells, which *lack* gap junctions, would *not* exhibit direct ion transfer between adjacent cells upon receipt of an action potential.

Educational objective:

The three main types of myocytes differ structurally. Skeletal and cardiac muscle cells have sarcomeres and are striated. In addition, cardiac and many smooth muscle cells have gap junctions, which allow these cells to directly transfer ions between adjacent cells.

Time spent:0 sec

Subject:
Biology

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System:
Musculoskeletal System